

ANSWERS

1.

- (a) probability of decay M1
of a nucleus per unit time A1
(allow 1 mark for $A = \lambda N$, with symbols explained)
- (b) (i) $\lambda = \ln 2 / (28 \times 365 \times 24 \times 3600)$ C1
 $= 7.85 \times 10^{-10} \text{ s}^{-1}$ A1
- (ii) $A = (-)\lambda N$
 $N = (6.4 \times 10^9) / (7.85 \times 10^{-10})$ C1
 $= 8.15 \times 10^{18}$ C1
mass = $(8.15 \times 10^{18} \times 90) / (6.02 \times 10^{23})$ (e.c.f. for value of N) C1
 $= 1.22 \times 10^{-3} \text{ g}$ A1
- (iii) volume = $(1.22 \times 10^{-3} / 2.54) = 4.8 \times 10^{-4} \text{ cm}^3$ A1
- (c) either very small volume of Strontium-90 has high activity
or dust can be highly radioactive B1
breathing in dust presents health hazard B1

2.

- (a) energy required to (completely) separate the nucleons (in a nucleus) B1
- (b) (i) U labelled near right-hand end of line B1
Ba and Kr in approximately correct positions B1
- (ii) binding energy is $A \times E_B$ B1
either binding energy of U < binding energy of (Ba + Kr)
or E_B of U < E_B of (Ba + Kr) B1
- (c) Krypton-92 reduced to 1/8 in 9 s M1
in 9 s, very little decay of Barium-141 M1
so, approximately 9 s A1
OR
 $\lambda_{\text{Kr}} = 0.231$ or $\lambda_{\text{Ba}} = 6.42 \times 10^{-4}$ (M1)
 $8 = e^{-\lambda_B \times t} / e^{-\lambda_K \times t}$ (C1)
 $t = 9.0 \text{ s}$ (A1)

3.

(a) (i) $\Delta N / \Delta t$ (ignore any sign) B1

(ii) $\Delta N / N$ (ignore any sign) B1

(b) source must decay by 8% C1

$$A = A_0 \exp(-\ln 2 \, t / T_{1/2}) \quad \text{or} \quad A/A_0 = 1 / (2^{t/T}) \quad \text{C1}$$

$$0.92 = \exp(-\ln 2 \times t / 5.27) \quad \text{or} \quad 0.92 = 1 / (2^{t/5.27}) \quad \text{C1}$$

$$t = 0.634 \text{ years} \quad \text{A1}$$

$$= 230 \text{ days}$$

(allow 2 marks for $A/A_0 = 0.08$, answer 7010 days)

allow 1 mark for $A/A_0 = 0.12$, answer 5880 days)

4.

(a) neutron is a single nucleon / particle B1

(b) binding energy = $4 \times 7.07 \times 1.6 \times 10^{-13}$ C1

$$= 4.52 \times 10^{-12} \text{ J}$$

$$\text{binding energy} = c^2 \Delta m \quad \text{C1}$$

$$4.52 \times 10^{-12} = (3.0 \times 10^8)^2 \times \Delta m$$

$$\Delta m = 5.03 \times 10^{-29} \text{ kg} \quad \text{A1}$$

(c) (i) fusion(do not allow fussion) B1

$$(ii) (2 \times 1.12) + 3x = 28.28 \quad \text{C1}$$

$$\dots -17.7 \quad \text{C1}$$

$$x = 2.78 \text{ MeV per nucleon} \quad \text{A1}$$

(use of +17.7 gives $x = 14.6 \text{ MeV}$, allow 1 mark only)

5.

(a) (constant) probability of decay M1

per unit time A1

(reference to decay of isotope / mass / sample / nuclide, allow max 1 mark)

(b) either when time = $t_{1/2}$, $N = \frac{1}{2}N_0$

$$\text{or} \quad \frac{1}{2}N_0 = N \exp(-\lambda t_{1/2}) \quad \text{C1}$$

$$\text{either} \quad 2 = \exp(\lambda t_{1/2})$$

$$\text{or} \quad \frac{1}{2} = \exp(-\lambda t_{1/2}) \quad \text{C1}$$

$$(\text{taking logs}), \ln 2 = 0.693 = \lambda t_{1/2} \quad \text{A1}$$

(c) $A = \lambda N$

$$1.8 \times 10^5 = N \times (0.693 / \{1.66 \times 10^{-27}\}) \quad \text{C1}$$

$$N = 4.3 \times 10^{13}$$

$$\text{mass} = 60 \times (N / N_A) \quad \text{or} \quad 60 \times N \times u \quad \text{C1}$$

$$= (60 \times 4.3 \times 10^{13}) / (6.02 \times 10^{23})$$

$$= 4.3 \times 10^{-9} \text{ g} \quad \text{A1}$$

6.

(a) (i) number = $(5.1 \times 10^{-6} \times 6.02 \times 10^{23}) / 241$	C1
= 1.27×10^{16}	A1

(ii) $A = \lambda N$	C1
$5.9 \times 10^5 = \lambda \times 1.27 \times 10^{16}$	
$\lambda = 4.65 \times 10^{-11} \text{ s}^{-1}$	A1

(iii) $4.65 \times 10^{-11} \times t_{1/2} = \ln 2$	C1
$t_{1/2} = 1.49 \times 10^{10} \text{ s}$	
= 470 years	A1

(b) sample / activity would decay appreciably whilst measurements are being made	B1
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7.

(a) (i) Fe shown near peak	A1
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(ii) Zr shown about half-way along plateau	A1
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(iii) H shown at less than 0.4 of maximum height	A1
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(b) (i) heavy / large nucleus breaks up / splits	M1
into two nuclei / fragments of approximately equal mass	A1

(ii) binding energy of nucleus = $B_E \times A$	B1
binding energy of parent nucleus is less than sum of binding energies of fragments	B1

8.

(a) energy required to separate nucleons in a <u>nucleus</u>	M1
to infinity	A1

(b) $1u = 1.66 \times 10^{-27} \text{ kg}$	
$E = mc^2$	C1
= $1.66 \times 10^{-27} \times (3.0 \times 10^8)^2$	M1
= $1.49 \times 10^{-10} \text{ J}$	
= $(1.49 \times 10^{-10}) / (1.6 \times 10^{-13})$	M1
= 930 MeV	A0

(c) (i) $\Delta m = 2.0141\text{u} - (1.0073 + 1.0087)\text{u}$
 $= -1.9 \times 10^{-3}\text{u}$ C1
 binding energy $= 1.9 \times 10^{-3} \times 930$
 $= 1.8\text{MeV}$ A1

(ii) $\Delta m = (57 \times 1.0087\text{u}) + (40 \times 1.0073\text{u}) - 97.0980\text{u}$ C1
 $= (-)0.69\text{u}$
 binding energy per nucleon $= (0.69 \times 930) / 97$ C1
 $= 6.61\text{MeV}$ A1

9.

(a) (i) time for initial number of nuclei/activity M1
 to reduce to one half of its initial value A1

(ii) $\lambda = \ln 2 / (24.8 \times 24 \times 3600)$ M1
 $= 3.23 \times 10^{-7} \text{ s}^{-1}$ A0

(b) (i) $A = \lambda N$ C1
 $3.76 \times 10^6 = 3.23 \times 10^{-7} \times N$
 $N = 1.15 \times 10^{13}$ A1

(ii) $N = N_0 e^{-\lambda t}$
 $= 1.15 \times 10^{13} \times \exp(-\{\ln 2 \times 30\} / 24.8)$ C1
 $= 4.97 \times 10^{12}$ A1

(c) ratio $= (4.97 \times 10^{12}) / (1.15 \times 10^{13} - 4.97 \times 10^{12})$ C1
 $= 0.76$ A1

10.

(a) (i) probability of decay (of a nucleus) M1
per unit time A1

(ii) $\lambda t_{1/2} = \ln 2$
 $\lambda = \ln 2 / (3.82 \times 24 \times 3600)$ M1
 $= 2.1 \times 10^{-6} \text{ s}^{-1}$ A0

(b) $A = \lambda N$ C1
 $200 = 2.1 \times 10^{-6} \times N$ C1
 $N = 9.5 \times 10^7$
 ratio $= (2.5 \times 10^{25}) / (9.5 \times 10^7)$
 $= 2.6 \times 10^{17}$ A1

11.

- (a) (i) *either* probability of decay (of a nucleus) per unit time B1
or $\lambda = (-)(dN/dt) / N$ A1
 $(-)(dN/dt)$ and N explained (M1)
(A1)
- (ii) in time $t_{1/2}$, number of nuclei changes from N_0 to $\frac{1}{2}N_0$ B1
 $\frac{1}{2} = \exp(-\lambda t_{1/2})$ or $2 = \exp(\lambda t_{1/2})$ B1
 $\ln(\frac{1}{2}) = -\lambda t_{1/2}$ and $\ln(\frac{1}{2}) = -0.693$ or $\ln 2 = \lambda t_{1/2}$ and $\ln 2 = 0.693$ B1
 $0.693 = \lambda t_{1/2}$ A0
- (b) $228 = 538 \exp(-8\lambda)$ C1
 $\lambda = 0.107 \text{ (hours}^{-1}\text{)}$ C1
 $t_{1/2} = 6.5 \text{ hours (do not allow 3 or more SF)}$ A1
- (c) e.g. random nature of decay
background radiation
daughter product is radioactive
(any two sensible suggestions, 1 each) B2

12.

- (a) nuclei having same number of protons/proton (atomic) number B1
different numbers of neutrons/neutron number B1
(allow second mark for nucleons/nucleon number/mass number/atomic mass if made clear that same number of protons/proton number)
- (b) probability of decay per unit time is the decay constant C1
 $\lambda = \ln 2 / t_{1/2}$
 $= 0.693 / (52 \times 24 \times 3600)$ C1
 $= 1.54 \times 10^{-7} \text{ s}^{-1}$ A1
- (c) (i) $A = A_0 \exp(-\lambda t)$
 $7.4 \times 10^6 = A_0 \exp(-1.54 \times 10^{-7} \times 21 \times 24 \times 3600)$ C1
 $A_0 = 9.8 \times 10^6 \text{ Bq}$ A1
(alternative method uses 21 days as 0.404 half-lives)
- (ii) $A = \lambda N$ and mass = $N \times 89 / N_A$ C1
mass = $(9.8 \times 10^6 \times 89) / (1.54 \times 10^{-7} \times 6.02 \times 10^{23})$
 $= 9.4 \times 10^{-9} \text{ g}$ A1

13.

- (a) (i) $x = 2$ A1
- (ii) *either* beta particle *or* electron B1
- (b) (i) mass of separate nucleons = $\{(92 \times 1.007) + (143 \times 1.009)\}$ u C1
 $= 236.931$ u C1
 binding energy = 236.931 u – 235.123 u
 $= 1.808$ u A1
- (ii) $E = mc^2$ C1
 energy = $1.808 \times 1.66 \times 10^{-27} \times (3.0 \times 10^8)^2$
 $= 2.7 \times 10^{-10}$ J C1
 binding energy per nucleon = $(2.7 \times 10^{-10}) / (235 \times 1.6 \times 10^{-13})$ M1
 $= 7.18$ MeV A0
- (c) energy released = $(95 \times 8.09) + (139 \times 7.92) - (235 \times 7.18)$ C1
 $= 1869.43 - 1687.3$
 $= 182$ MeV A1
(allow calculation using mass difference between products and reactants)

14.

- (a) two (light) nuclei combine M1
 to form a more massive nucleus A1
- (b) (i) $\Delta m = (2.01410 \text{ u} + 1.00728 \text{ u}) - 3.01605 \text{ u}$
 $= 5.33 \times 10^{-3} \text{ u}$ C1
 energy = $c^2 \times \Delta m$ C1
 $= 5.33 \times 10^{-3} \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$
 $= 8.0 \times 10^{-13} \text{ J}$ A1
- (ii) speed/kinetic energy of proton and deuterium must be very large B1
 so that the nuclei can overcome electrostatic repulsion B1

15.

- (a) energy is given out / released on formation of the α -particle (or reverse argument) M1
 either $E = mc^2$ so mass is less
 or reference to mass-energy equivalence A1
- (b) (i) mass change = $18.00567 \text{ u} - 18.00641 \text{ u}$ C1
 $= 7.4 \times 10^{-4} \text{ u}$ (*sign not required*) A1

(ii) energy = $c^2 \Delta m$
 $= (3.0 \times 10^8)^2 \times 7.4 \times 10^{-4} \times 1.66 \times 10^{-27}$ C1
 $= 1.1 \times 10^{-13} \text{ J}$ A1
 (allow use of $u = 1.67 \times 10^{-27} \text{ kg}$)
 (allow method based on 1u equivalent to 930 MeV to 933 MeV)

- (iii) either mass of products greater than mass of reactants M1
 this mass/energy provided as kinetic energy of the helium-4 nucleus A1
 or both nuclei positively charged (M1)
 energy required to overcome electrostatic repulsion (A1)

16.

- (a) energy to separate nucleons (in a nucleus) M1
 separate to infinity A1

- (b) (i) fission B1

- (ii) 1. U: near right-hand end of line B1

2. Mo: to right of peak, less than 1/3 distance from peak to U B1

3. La: 0.4 $\rightarrow$ 0.6 of distance from peak to U B1

- (iii) 1. right-hand side, mass = 235.922 u C1
 mass change = 0.210 u A1

2. energy = mc^2 C1
 $= 0.210 \times 1.66 \times 10^{-27} \times (3.0 \times 10^8)^2$
 $= 3.1374 \times 10^{-11} \text{ J}$ C1
 $= 196 \text{ MeV}$ (need 3 s.f.) A1

(use of 1 u = 934 MeV, allow 3/3; use of 1 u = 930 MeV or 932 MeV, allow 2/3)

(use of 1.67×10^{-27} not 1.66×10^{-27} scores max. 2/3)

17.

- (a) probability of decay (of a nucleus)/fraction of number of nuclei in sample that decay M1
 per unit time A1
 (allow $\lambda = (dN / dt) / N$ with symbols explained – (M1), (A1))

- (b) (i) number = $(1.2 \times 6.02 \times 10^{23}) / 235$ C1
 $= 3.1 \times 10^{21}$ A1

- (ii) $N = N_0 e^{-\lambda t}$
 negligible activity from the krypton B1
 for barium, $N = (3.1 \times 10^{21}) \exp(-6.4 \times 10^{-4} \times 3600)$
 $= 3.1 \times 10^{20}$ C1
 activity $= \lambda N$
 $= 6.4 \times 10^{-4} \times 3.1 \times 10^{20}$ C1
 $= 2.0 \times 10^{17} \text{ Bq}$ A1

18.

- (a) energy required to separate the nucleons (in a nucleus) M1
 to infinity A1
 (allow reverse statement)

- (b) (i) $\Delta m = (2 \times 1.00867) + 1.00728 - 3.01551$ C1
 $= 9.11 \times 10^{-3} \text{ u}$ C1
 binding energy $= 9.11 \times 10^{-3} \times 930$
 $= 8.47 \text{ MeV}$ A1
 (allow 930 to 934 MeV so answer could be in range 8.47 to 8.51 MeV)
 (allow 2 s.f.)

- (ii) $\Delta m = 211.70394 - 209.93722$
 $= 1.76672 \text{ u}$ C1
 binding energy per nucleon $= (1.76672 \times 930)/210$ C1
 $= 7.82 \text{ MeV}$ A1
 (allow 930 to 934 MeV so answer could be in range 7.82 to 7.86 MeV)
 (allow 2 s.f.)

- (c) total binding energy of barium and krypton M1
 is greater than binding energy of uranium A1

19.

- (a) time for number of atoms / nuclei / activity (of the isotope) M1
 to be reduced to one half (of its initial value) A1

- (b) (i) $A = \lambda N$ C1
 $460 = N \times \ln 2 / (8.1 \times 24 \times 60 \times 60)$ C1
 $N = 4.6 \times 10^8$ A1

- (ii) number of water molecules in 1.0 kg $= (6.02 \times 10^{23}) / (18 \times 10^{-3})$ C1
 $= 3.3 \times 10^{25}$
 ratio $= (3.3 \times 10^{25}) / (4.6 \times 10^8)$
 $= 7.2 (7.3) \times 10^{16}$ A1

- (c) $A = A_0 e^{-\lambda t}$ and $\lambda t_{1/2} = \ln 2$ C1
 $170 = 460 \exp(-\{\ln 2\} / 8.1)$ C1
 $t = 11.6 \text{ days}$ (allow 2 s.f.) A1

20.

(a) 'light' nuclei combine to form 'heavier' nuclei

B1

(b) (i) *either* $\text{energy} = c^2 \Delta m$

or $\text{energy} = (3.00 \times 10^8)^2 \times 1.66 \times 10^{-27}$

C1

$\text{energy} = 1.494 \times 10^{-10} \text{ J}$

C1

$= (1.494 \times 10^{-10}) / (1.60 \times 10^{-13})$

$= 934 \text{ MeV (3 s.f.)}$

A1

(ii) $\Delta m = (2.01356 + 3.01551) - (4.00151 + 1.00867)$

$= 5.02907 - 5.01018$

$= 0.01889 \text{ u}$

C1

$\text{energy} = 0.01889 \times 934$

$= 17.6 \text{ MeV (allow 2 s.f.)}$

A1

(iii) high temperature means high speeds / kinetic energy of nuclei

B1

D and T nuclei collide despite repelling one another

B1

21.

(a) $\text{activity} = (1.7 \times 10^{14}) / (2.5 \times 10^6)$

$= 6.8 \times 10^7 \text{ Bq kg}^{-1}$

A1

(b) (i) energy released per second in 1.0 kg of steel

$= 6.8 \times 10^7 \times 0.067 \times 1.6 \times 10^{-13}$

$= 7.3 \times 10^{-7} \text{ J}$

B1

(ii) this is a very small quantity of energy so steel will not be warm

B1

(iii) $A = A_0 e^{-\lambda t}$ and $\lambda t_{1/2} = \ln 2$

C1

$400 = (6.8 \times 10^7) \exp(-[\ln 2 \times t] / 92)$

C1

$t = 1600 \text{ years}$

A1

or

$A = A_0 / 2^n$

(C1)

$n = 17.4$

(C1)

$t = 17.4 \times 92 = 1600 \text{ years}$

(A1)

22.

- (a) nucleus/nuclei emits M1
 spontaneously/randomly A1
 α -particles, β -particles, γ -ray photons A1
- (b) (i) $N - \Delta N$ A1
 (ii) $\Delta N / \Delta t$ A1
 (iii) $\Delta N / N$ A1
 (iv) $\Delta N / N \Delta t$ A1
- (c) graph: smooth curve in correct direction starting at (0,0) M1
 n at $2t_{1/2}$ is 1.5 times that at $t_{1/2}$ (± 2 mm) A1

23.

- (a) (i) energy = $5.75 \times 1.6 \times 10^{-13}$
 $= 9.2 \times 10^{-13} \text{ J}$ A1
- (ii) number = $1900 / (9.2 \times 10^{-13} \times 0.24)$ C1
 $= 8.6 \times 10^{15} \text{ s}^{-1}$ A1
- (b) (i) decay constant = $0.693 / (2.8 \times 365 \times 24 \times 3600)$ C1
 $= 7.85 \times 10^{-9} \text{ s}^{-1}$ (allow 7.8 or 7.9 to 2 s.f.) A1
- (ii) $A = \lambda N$
 $8.6 \times 10^{15} = 7.85 \times 10^{-9} \times N$ C1
 $N = 1.096 \times 10^{24}$ C1
- mass = $(1.096 \times 10^{24} \times 236) / (6.02 \times 10^{23})$ M1
 $= 430 \text{ g}$ A1
- (c) $0.84 = 1.9 \exp(-7.85 \times 10^{-9} t)$ C1
 $t = 1.04 \times 10^8 \text{ s}$
 $= 3.3 \text{ years}$ A1

24.

- (a) nucleus: small central part/core of an atom B1
 nucleon: proton or a neutron B1
 particle contained within a nucleus B1

(b) (i) 1. decay constant $= \ln 2 / (3.8 \times 24 \times 3600)$ C1
 $= 2.1 \times 10^{-6} \text{ s}^{-1}$ A1

2. $A = \lambda N$
 $97 = 2.1 \times 10^{-6} \times N$ C1
 $N = 4.6 \times 10^7$ A1

(ii) 1.0 m^3 contains $(6.02 \times 10^{23}) / (2.5 \times 10^{-2})$ air molecules C1

ratio $= (4.6 \times 10^7 \times 2.5 \times 10^{-2}) / (6.02 \times 10^{23})$
 $= 1.9 \times 10^{-18}$ A1

25.

- (a) energy required to separate the nucleons (in a nucleus)
 or energy required to separate the protons and neutrons in a nucleus M1
 (or energy released when nucleons combine (to form a nucleus)/energy released
 when protons and neutrons combine to form a nucleus)

either completely or to infinity A1
 (either free protons and neutrons or from infinity)

- (b) (i) either different forms of same element or nuclei having same number of
 protons with different numbers of neutrons M1
 A1

(ii) 1784 MeV (accept min. 3 s.f.) A1
 7.57 MeV A1

(c) (i) $\lambda = \ln 2 / (7.1 \times 10^8 \times 365 \times 24 \times 3600) = 3.1 \times 10^{-17} \text{ s}^{-1}$ B1

(ii) $A = \lambda N$
 $5000 = 3.1 \times 10^{-17} \times N$ C1
 $N = 1.61 \times 10^{20}$

mass $= 235 \times (1.61 \times 10^{20}) / (6.02 \times 10^{23})$ C1
 $= 0.063 \text{ g}$ (accept min. 2 s.f.) A1

26.

(a) $\lambda = \ln 2 / T_{1/2}$
 $= \ln 2 / (53.3 \times 24 \times 60 \times 60) = 1.5 \times 10^{-7} \text{ s}^{-1}$ A1

(b) $A = \lambda N$ C1

$$N = 39 \times 10^{-3} / 1.5 \times 10^{-7} = 2.6 \times 10^5$$

$$m = (2.6 \times 10^5 / 6.0 \times 10^{23}) \times 7 \times 10^{-3} \text{ or } 2.6 \times 10^5 \times 1.66 \times 10^{-27} \times 7$$
 C1

$$= 3.0 \times 10^{-21} \text{ kg}$$
 A1

(c) $2/39 = \exp(-1.5 \times 10^{-7} \times t)$ or $2/39 = (1/2)^{[t/(53.3 \times 24 \times 3600)]}$ C1

$$t = 2.0 \times 10^7 \text{ s}$$
 A1

27.

(a) e.g. background count (rate)/radiation

multiple possible counts from each decay

radiation emitted in all directions

dead-time of counter

(daughter) product unstable/also emits radiation

self-absorption of radiation in sample or absorption in air/detector window

three sensible suggestions, 1 each B3

(b) $A = A_0 \exp(-\ln 2 \times t / T_{1/2})$

$$1.21 \times 10^2 = 3.62 \times 10^4 \exp(-\ln 2 \times 42.0 / T_{1/2})$$

or

$$1.21 \times 10^2 = 3.62 \times 10^4 \exp(-\lambda \times 42.0)$$
 C1

$$T_{1/2} = 5.1 \text{ minutes (306 s)}$$
 A1

(c) discrete energy levels (in nuclei) B1

28.

- (a) (photons of) electromagnetic radiation M1
emitted from nuclei A1
- (b) line of best fit drawn B1
recognises μ as given by the gradient of best-fit line
or
 $\ln C = \ln C_0 - \mu x$ B1
 $\mu = 0.061 \text{ mm}^{-1}$ (within $\pm 0.004 \text{ mm}^{-1}$, 1 mark; within $\pm 0.002 \text{ mm}^{-1}$, 2 marks) A2
- (c) aluminium is less absorbing (than lead)
or
gradient of graph would be less M1
so μ is smaller A1

29.

- (a) (i) time for number of atoms/nuclei or activity to be reduced to one half M1
reference to (number of...) original nuclide/single isotope
or
reference to half of original value/initial activity A1
- (ii) $A = A_0 \exp(-\lambda t)$ and either $t = t_{1/2}$, $A = \frac{1}{2}A_0$ or $\frac{1}{2}A_0 = A_0 \exp(-\lambda t_{1/2})$ M1
so $\ln 2 = \lambda t_{1/2}$ (and $\ln 2 = 0.693$), hence $0.693 = \lambda t_{1/2}$ A1
- (b) $A = \lambda N$
 $N = 200 / (2.1 \times 10^{-6})$ C1
 $= 9.52 \times 10^7$ C1
mass = $(9.52 \times 10^7 \times 222 \times 10^{-3}) / (6.02 \times 10^{23})$
or
mass = $9.52 \times 10^7 \times 222 \times 1.66 \times 10^{-27}$ C1
 $= 3.5 \times 10^{-17} \text{ kg}$ A1

30.

(a) (i) ${}^4_2\text{He}$ or ${}^4_2\alpha$ B1

(ii) ${}^1_0\text{n}$ B1

(b) (i) $\Delta m = (29.97830 + 1.00867) - (26.98153 + 4.00260)$ C1

$$= 30.98697 - 30.98413$$

$$= 2.84 \times 10^{-3} \text{ u} \quad \text{C1}$$

(ii) $E = c^2\Delta m$ or mc^2 C1

$$= (3.0 \times 10^8)^2 \times 2.84 \times 10^{-3} \times 1.66 \times 10^{-27}$$

$$= 4.2 \times 10^{-13} \text{ J} \quad \text{A1}$$

(c) mass of products is greater than mass of Al plus α

or

reaction causes (net) increase in (rest) mass (of the system) B1

α -particle must have at least this amount of kinetic energy B1

31.

12(a)	either	
	(minimum) energy required/work done to separate the nucleons (in a nucleus)	M1
	to infinity	A1
	or	
	energy released when nucleons come together (to form a nucleus)	(M1)
	from infinity	(A1)
12(b)(i)	(total) binding energy of thorium and helium (nuclei) greater than binding energy of uranium (nucleus)	B1
12(b)(ii)1	change in mass = $238.05076 - (234.04357 + 4.00260)$ = $4.59 \times 10^{-3} \text{ u}$	A1
12(b)(ii)2	either	
	$E = mc^2$	C1
	= $4.59 \times 10^{-3} \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$	
	= $6.9 \times 10^{-13} \text{ J}$	A1
	or	
	1u = 931 MeV $E = 4.59 \times 10^{-3} \times 931 \times 10^6 \times 1.6 \times 10^{-19}$	(C1)
12(b)(iii)	Th nucleus / He nucleus / product nucleus has kinetic energy	M1
	energy of gamma photon must be less than energy released	A1

32.

12(a)	$x = 7$	A1
12(b)(i)	$E = mc^2$	C1
	$= 1.66 \times 10^{-27} \times (3.0 \times 10^8)^2$	C1
	$= 1.494 \times 10^{-10} \text{ J}$	
	division by 1.6×10^{-13} clear to give 934 MeV	A1
12(b)(ii)	$\Delta m = (235.123 + 1.00863) - (94.945 + 138.955 + 2 \times 1.00863 + 7 \times 5.49 \times 10^{-4})$ or $\Delta m = 235.123 - (94.945 + 138.955 + 1 \times 1.00863 + 7 \times 5.49 \times 10^{-4})$	C1
	$= 0.21053 \text{ u}$	C1
	energy $= 0.21053 \times 934$	A1
	$= 197 \text{ MeV}$	
12(c)	kinetic energy of nuclei/particles/products/fragments	B1
	γ -ray photon energy	B1

33.

12(a)	$7 \text{ }^0_{-1}\text{e}$	A1
12(b)(i)	$E = mc^2$	C1
	$= 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$	M1
	$= 1.494 \times 10^{-10} \text{ J}$	A1
	division by 1.60×10^{-13} clear to give 934 MeV	
12(b)(ii)	$\Delta m = (82 \times 1.00863\text{u}) + (57 \times 1.00728\text{u}) - 138.955\text{u}$	C1
	$= (-) 1.16762 \text{ (u)}$	
	energy $= 1.16762 \times 934$	C1
	energy per nucleon $= (1.16762 \times 934) / 139$ $= 7.85 \text{ MeV}$	A1
12(c)	above $A = 56$, binding energy per nucleon decreases as A increases	B1
	U-235 has larger nucleon number	M1
	so less (binding energy per nucleon)	A1
	or	
	fission takes place with uranium	(B1)
	fission reaction releases energy	(M1)
	binding energy per nucleon less (for uranium than for products)	(A1)

34.

12(a)	• emission from radioactive daughter products • self-absorption in source • absorption in air before reaching detector • detector not sensitive to all radiations • window of detector may absorb some radiation • dead-time of counter • background radiation Any two points.	B2
12(b)(i)	curve is not smooth or curve fluctuates/curve is jagged	B1
12(b)(ii)	clear evidence of allowance for background	B1
	half-life determined at least twice	B1
	half-life = 1.5 hours (1 mark if in range 1.7–2.0; 2 marks if in range 1.4–1.6)	A2
12(c)	1. half-life: no change	M1
	because decay is spontaneous/independent of environment	A1
	2. count rate (likely to be or could be) different/is random/cannot be predicted	B1

35.

12(a)(i)	nucleus emits particles/EM radiation/ionising radiation	B1
	emission/release from unstable nucleus or emission from nucleus is random and/or spontaneous	B1
12(a)(ii)	probability of decay (of a nucleus) or fraction of (number of undecayed) nuclei that will decay	M1
	per unit time	A1
12(b)	energy is shared with another particle	B1
	mention of antineutrino	B1
12(c)(i)	number = $[(1.2 \times 10^{-9}) / 131] \times 6.02 \times 10^{23}$ or number = $(1.2 \times 10^{-3} \times 10^{-9}) / (131 \times 1.66 \times 10^{-27})$ (= 5.51×10^{12})	C1
	$A = \lambda N$	C1
	= $[0.086 / (24 \times 3600)] \times 5.51 \times 10^{12}$ = 5.5×10^6 Bq	A1
12(c)(ii)	$1/50 = \exp(-0.086t)$ or $1/50 = 0.5^n$	C1
	$t = 45$ days	A1

36.

13(a)(i)	probability of decay (of a nucleus)	M1
	per unit time	A1
13(a)(ii)	$A = A_0 e^{-\lambda t}$	M1
	after one half-life, $\frac{1}{2}A_0 = A_0 e^{-\lambda t_{1/2}}$	
	$\frac{1}{2} = \exp(-\lambda t_{1/2})$ and hence taking logs, $\ln 2 = \lambda t_{1/2}$	A1
13(b)	activity = $3.8 \times 10^4 \exp(-\ln 2 \times 36 / 15)$	C1
	= 7200 Bq	C1
	or activity = $3.8 \times 10^4 / 2^{2.4}$	(C1)
	= 7200 Bq	(C1)
	volume = $(7200 / 1.2) \times 5.0$	C1
	= $3.0 \times 10^4 \text{ cm}^3$	A1
	OR activity of $5.0 \text{ cm}^3 = 1.2 \times 2^{2.4}$	(C1)
	= 6.3336 Bq	(C1)
	volume = $(3.8 \times 10^4 / 6.3336) \times 5.0$	(C1)
	= $3.0 \times 10^4 \text{ cm}^3$	(A1)

37.

13(a)	emission of particles/radiation by unstable nucleus	B1
	spontaneous emission	B1
13(b)(i)	use of graph to determine half-life = 14 minutes	B1
	hence $\lambda = \ln 2 / (14 \times 60) (\text{s}^{-1})$	C1
	N at 14 minutes = 4.4×10^7 and $A = \lambda N$	C1
	activity = $4.4 \times 10^7 \times \ln 2 / (14 \times 60)$	A1
	= $3.6 \times 10^4 \text{ Bq}$	
	or	
	correct tangent drawn at time $t = 14$ minutes	(B1)
	magnitude of gradient of tangent identified as activity	(C1)
	correct working for gradient leading to activity	(C1)
	activity = $3.6 \times 10^4 \text{ Bq}$	(A1)
13(b)(ii)	$3.6 \times 10^4 = \lambda \times 4.4 \times 10^7$	C1
	or $\lambda = \ln 2 / (14.0 \times 60)$	
	$\lambda = 8.2 \times 10^{-4} \text{ s}^{-1}$	A1

38.

12(a)	emission of particles/radiation by <u>unstable nucleus</u>	B1
	spontaneous emission	B1
12(b)(i)	P – the curve that starts with a high number D – the curve with the peak S – the curve that increases from zero throughout (one correct 1 mark, all three correct 2 marks)	B2
12(b)(ii)	$\lambda t_{1/2} = 0.693$	C1
	$\lambda = 0.693 / (60.0 \times 60)$	
	$= 1.93 \times 10^{-4} \text{ s}^{-1}$	A1
12(c)	half-life of F is much shorter than half-life of E	B1
	<u>nuclei</u> of F decay (almost) as soon as they are produced	B1

39.

12(a)	unstable nucleus	B1
	emission of particles/photons	B1
	emission is spontaneous or (particles/radiation) are ionising	B1
12(b)(i)	tangent drawn and gradient calculation attempted	B1
	activity = $1.3 \times 10^6 \text{ Bq}$ (1 mark for answer within $\pm 0.2 \times 10^6 \text{ Bq}$, 2 marks for answer within $\pm 0.1 \times 10^6 \text{ Bq}$)	A2
12(b)(ii)	$A = \lambda N$	C1
	$\lambda = (1.3 \times 10^6) / (3.05 \times 10^{10}) = 4.3 \times 10^{-5} \text{ s}^{-1} (\approx 4 \times 10^{-5} \text{ s}^{-1})$	A1
12(c)	$A = A_0 e^{-\lambda t}$	C1
	$1.0 \times 10^3 = 4.6 \times 10^3 \exp(-5.5 \times 10^{-7} \times t)$	
	$\ln(4.6) = 5.5 \times 10^{-7} \times t$	C1
	$t = 2.78 \times 10^6 \text{ s}$ $= 32 \text{ days}$	A1

40.

12(a)	fusion: two nuclei <u>combine</u> to form a (single) nucleus	B1
	fission: a (single) large nucleus <u>divides</u> to form (smaller) nuclei	B1
	Any one from: - fusion is initiated by (very) high temperatures - fission is initiated by neutron bombardment - resulting nuclei in fission are of similar size (both processes) release energy - binding energy per nucleon increases - total binding energy increases - fission involves release of neutrons	B1
12(b)(i)	neutron	B1
12(b)(ii)	1. zero	A1
	2. $(4 \times 11.3290 \times 10^{-13}) - (2 \times 1.7813 \times 10^{-13}) - (3 \times 4.5285 \times 10^{-13})$	C1
	energy change = $45.316 \times 10^{-13} - 17.148 \times 10^{-13}$ $= 2.82 \times 10^{-12} \text{ J}$	A1

12(b)(iii)	1.0 mol or N_A nuclei of each energy = $2.817 \times 10^{-12} \times 6.02 \times 10^{23}$ = 1.7×10^{12} J	A1
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41.

12(a)(i)	fission	B1
12(a)(ii)	either ${}^0_{-1}\text{e}$ or ${}^0_{-1}\beta$	M1
	7	A1
12(b)(i)	energy = $c^2 \Delta m$ = $0.223 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$	C1
	= 3.33×10^{-11} J	A1
12(b)(ii)	Any 2 from: kinetic energy of products gamma photons neutrinos	B2

42.

12(a)(i)	$\Delta N / \Delta T$	B1
12(a)(ii)	$\Delta N / N$	B1
12(a)(iii)	$\Delta N / (N \Delta T)$	B1
12(b)(i)	1. mass change = 5.60×10^{-3} u	A1
	2. energy = $(\Delta)mc^2$	C1
	= $5.6 \times 10^{-3} \times 1.66 \times 10^{-27} \times (3.0 \times 10^8)^2$ (= 8.36×10^{-13} J)	C1
	= 0.84 pJ	A1
12(b)(ii)	kinetic energy (of recoil) of lead (nucleus)	B1
	energy of γ -ray photon	B1

43.

12(a)	energy required to separate the nucleons (in a nucleus)	M1
	to infinity	A1
	or	
	energy released when nucleons come together (to form nucleus)	(M1)
	from infinity	(A1)
12(b)	mass defect = $140.911 - (57 \times 1.007) - (84 \times 1.009)$	C1
	= $140.911 - 142.155$ = (-1.244) (u)	C1
	energy = $c^2(\Delta)m$	C1
	= $(3.00 \times 10^8)^2 \times 1.244 \times 1.66 \times 10^{-27}$ = 1.9×10^{-10} J	A1

12(c)(i)	$A = A_0 e^{-\lambda t}$ and $\ln 2 = \lambda t_{1/2}$	C1
	$0.40 = \exp(-\ln 2 \times t / 3.9)$	C1
	or	
	$(0.5)^n = 0.40$	(C1)
	$n = 1.32$ and $t = 1.32 \times 3.9$	(C1)
	$t = 5.2$ hours	A1
12(c)(ii)	daughter product may be radioactive or random nature of decay	B1

44.

12(a)	$E = (3.0 \times 10^8)^2 \times 1.66 \times 10^{-27} (= 1.49 \times 10^{-10} \text{ J})$	M1
	$= (1.49 \times 10^{-10}) / (1.60 \times 10^{-19}) = 9.34 \times 10^8 = 934 \text{ MeV}$	A1
	or	
	binding energy = 8.443×95 [or equivalent using La-139 nucleus]	(M1)
	binding energy / mass defect = $(8.443 \times 95) / 0.859 = 934 \text{ MeV}$	(A1)
12(b)(i)	binding energy = $1.865 \times 934 (= 1741.91 \text{ MeV})$	C1
	binding energy per nucleon = $1741.91 / 235$ $= 7.41 \text{ (MeV)}$	A1
12(b)(ii)	less (than)	B1
12(c)	energy = $\{(1.219 + 0.859) - 1.865\} \times 934$ or energy = $(95 \times 8.443) + (139 \times 8.189) - (235 \times 7.412)$	C1
	$= 199 \text{ MeV}$	A1
12(d)	number of reactions = $1.2 \times 10^{-7} \times 6.02 \times 10^{23}$ $= 7.22 \times 10^{16}$	C1
	energy release (for one reaction) = $199 \times 1.60 \times 10^{-13} (= 3.18 \times 10^{-11} \text{ J})$	C1
	power = $(7.22 \times 10^{16} \times 3.18 \times 10^{-11}) / (25 \times 10^{-3})$ $= 9.2 \times 10^7 \text{ W}$	A1

45.

12(a)(i)	(decay is) unpredictable/cannot be predicted	B1
12(a)(ii)	probability of decay (of a nucleus)	M1
	per unit time	A1
12(b)	$A = \lambda N$	C1
	(for 1.00 m^3) $A = 0.600 / 4.80 \times 10^{-3} (= 125 \text{ Bq})$	C1
	$N = 125 / ([7.55 \times 10^{-3}] / 3600) (= 5.96 \times 10^7)$	C1
	so ratio = $(2.52 \times 10^{25}) / (5.96 \times 10^7)$	C1
	or	
	(for $4.80 \times 10^{-3} \text{ m}^3$) N for air = $2.52 \times 10^{25} \times 4.80 \times 10^{-3} (= 1.21 \times 10^{23})$	(C1)
	N for radon = $0.600 / ([7.55 \times 10^{-3}] / 3600) (= 2.86 \times 10^5)$	(C1)
	so ratio = $(1.21 \times 10^{23}) / (2.86 \times 10^5)$	(C1)
	ratio = 4.2×10^{17}	A1

46.

12(a)	(minimum) energy required to separate the nucleons	M1
	to infinity	A1
12(b)(i)	${}_{37}^{137}\text{Ba}$	B1
12(b)(ii)	fission	B1
12(b)(iii)	binding energy per nucleon smaller for U than for Cs	B1
12(c)	Current ratio 2 Y to 1 Zr, so initially 3 Y $2 = 3 e^{-\lambda t}$ $\lambda = 0.693 / 2.7$ $\ln(2/3) = -(\ln 2 / 2.7)t$	C1
	$t = 1.6$ days	A1
	or	
	$(\frac{1}{2})^n = 2/3$	(C1)
	$n = 0.585$	(C1)
	time = 0.585×2.7 = 1.6 days	(A1)

47.

12(a)(i)	time at which a nucleus will decay cannot be predicted or constant probability of decay of a nucleus	B1
12(a)(ii)	decay (of a nucleus) not affected by environmental factors	B1
12(b)	$A = A_0 e^{-\lambda t}$ and $\lambda = \ln 2 / t_{1/2}$	C1
	$= 3.6 \times 10^5 \times \exp [-(2 \times \ln 2) / 1.4]$	C1
	or	
	$A = A_0 \times 0.5^N$	(C1)
	$= 3.6 \times 10^5 \times 0.5^N$ where $N = 2 / 1.4$	(C1)
	$A = 1.3 \times 10^5$ Bq	A1
12(c)(i)	smooth curve, starting at (0, 3.6×10^5) and passing through (1.4, 1.8×10^5) and (2.0, 1.3×10^5)	B1
12(c)(ii)	(activity of sample is greater than activity of X so) there must be an additional source of activity	C1
	the decay product (of isotope X) is radioactive	A1

48.

12(a)	difference between mass of nucleus and mass of (constituent) nucleons	M1
	where nucleons are separated to infinity	A1
12(b)(i)	$E = mc^2$	C1
	$= 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2 / (1.60 \times 10^{-13}) = 934$ MeV	A1
12(b)(ii)	mass defect = $2 \times (1.007276 + 1.008665) - 4.001506$ (= 0.030376)	B1
	binding energy per nucleon = $(0.030376 \times 934) / 4 = 7.09$ MeV	A1
12(c)	binding energy per nucleon is much greater	M1
	so would require a large amount of energy to separate the nucleons in helium	A1

	or	
	amount of energy released in forming hydrogen isotopes	(M1)
	is less than energy required to break apart helium nucleus	(A1)

49.

12(a)(i)	unstable nucleus	B1
	emits ionising radiation or decays spontaneously	B1
12(a)(ii)	probability of decay (of a nucleus)	M1
	per unit time	A1
12(b)	$A = \lambda N$	C1
	$560 = 9.9 \times 10^{-7} \times N$	A1
	$N = 5.7 \times 10^8$	
12(c)	$A = A_0 e^{-\lambda t}$	C1
	$170 = 560 \exp(-9.9 \times 10^{-7} \times t)$	
	$t = 1.2 \times 10^6 \text{ s}$	C1
	$= 14 \text{ days}$	A1

50.

12(a)(i)	energy required to separate nucleons (of nucleus)	M1
	to infinity	A1
12(a)(ii)	a (single) large nucleus <u>divides</u> to form (smaller) nuclei	B1
	any one point from: - initiated by neutron bombardment - resulting nuclei are of similar size - binding energy per nucleon increases - total binding energy increases - neutrons released - combined mass of smaller nuclei is less than mass of large nucleus	B1
12(b)	binding energy per nucleon is a maximum at around $A = 56$	B1
	products of splitting a ^{56}Fe nucleus must have a lower total binding energy	B1
	(reaction would require) a net input of energy	B1

51.

12(a)	1 not affected by external factors	B1
	2 cannot predict when a (particular) nucleus will decay or cannot predict which nucleus will decay (next)	B1
12(b)(i)	Number of atoms = $\frac{1.0 \times 10^{-9}}{90 \times 1.66 \times 10^{-27}}$ or $\frac{1.0 \times 10^{-9} \times 6.02 \times 10^{23}}{90 \times 10^{-3}}$	C1
	$= 6.693 \times 10^{15}$	
	$A = \lambda N$	C1
	$\lambda = \frac{5.2 \times 10^6}{6.693 \times 10^{15}}$	
	$\lambda = 7.8 \times 10^{-10} \text{ s}^{-1}$	A1
12(b)(ii)	daughter nucleus is unstable	B1

52.

12(a)	probability of decay (of a nucleus)	M1
	per unit time	A1
12(b)	$A = \lambda N$	C1
	$N = \text{mass} / (\text{nucleon number} \times u)$	C1
	$2.92 \times 10^9 = (\lambda \times 5.87 \times 10^{-10}) / (131 \times 1.66 \times 10^{-27})$	A1
	$\lambda = 1.08 \times 10^{-6} \text{ s}^{-1}$	
12(c)	- sample emits radiation in all directions - some radiation is absorbed by air/detector window - self-absorption within the source - dead time/inefficiency of detector Any two points, 1 mark each	B2

53.

12(a)(i)	cannot predict when a particular nucleus will decay or cannot predict which nucleus will decay next	B1
12(a)(ii)	(decay is) not affected by external (environmental) factors	B1
12(b)(i)	$A = A_0 \exp(-\lambda t)$ and so $\ln A = \ln A_0 - \lambda t$	C1
	gradient of line = $(-\lambda)$	
	$\lambda = (36.4 - 35.0) / (20 - 0)$	C1
	$(= 0.07(0) \text{ min}^{-1})$	
	half-life = $\ln 2 / \lambda$	A1
	$= \ln 2 / 0.070$	
	$= 10 \text{ min}$	
	or	
	$A_0 = \exp(-36.4) = 6.43 \times 10^{15} \text{ (Bq)}$	(C1)
	$A_0 / 2 = 3.21 \times 10^{15} \text{ (Bq)}$, so $\ln(A_0 / 2) = 35.7$	(C1)
	read off half-life = 10 min	(A1)
	or	
12(b)(ii)	(at one half-life,) $\ln A = 36.4 - \ln 2$	(C1)
	$= 35.7$	(C1)
	read off half-life = 10 min	(A1)
	$A = \lambda N$	C1
	$N = \text{mass} / (\text{nucleon number} \times u)$ or $N = (\text{mass} / \text{nucleon number}) \times N_A$	C1
	$\exp(36.4) = (1.17 \times 10^{-3} \times 5.66 \times 10^{-7}) / (\text{nucleon number} \times 1.66 \times 10^{-27})$ or $\exp(36.4) = (1.17 \times 10^{-3} \times 5.66 \times 10^{-4} \times 6.02 \times 10^{23}) / \text{nucleon number}$ nucleon number = 62	A1

54.

9(a)	207, 82 for lead	B1
	4, 2 for alpha	B1
9(b)(i)	(half-life found as) 0.52 s or correctly read points substituted into $N = N_0 e^{-\lambda t}$ $\lambda = \frac{0.693}{t_{1/2}}$ $\lambda = \frac{0.693}{0.52}$	C1
	$\lambda = 1.3 \text{ s}^{-1}$	A1
9(b)(ii)	$A = \lambda N$ $= 1.3 \times 24 \times 10^{12}$ $= 3.1 \times 10^{13} \text{ Bq}$	A1
9(b)(iii)	upwards curve of decreasing gradient starting from (0,0)	B1
	passes through (0.52, 12) and (1.2, 18.8)	B1
9(c)(i)	$16 \times 10^{12} \text{ and } 7.2 \times 10^{12}$	C1
	$6900 \times 10^3 \times 1.6 \times 10^{-19}$	C1
	$(16 \times 10^{12} - 7.2 \times 10^{12}) \times 6900 \times 10^3 \times 1.6 \times 10^{-19}$	
	$= 9.7 \text{ J}$	A1
9(c)(ii)	lead nuclei have kinetic energy	B1
	or	
	gamma <u>photons</u> are also emitted	

55.

8(a)(i)	energy required to separate the nucleons (in the nucleus)	M1
	to infinity	A1
8(a)(ii)	curve starting close to the origin and forming a single peak	B1
	peak shown to left of centre, with steep line on LHS of peak and shallow line on RHS of peak	B1
8(b)(i)	fusion	B1
8(b)(ii)	both particles have low A values or both particles are at left-hand end of graph	B1
	He-3 has higher binding energy (per nucleon) than H-2	B1
8(c)	$\Delta m = [(2 \times 2.014102) - (3.016029 + 1.008665)] \text{ u}$ $(= 0.00351 \text{ u})$	C1
	$E = \Delta mc^2$	C1
	$= 0.00351 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$ $(= 5.24 \times 10^{-13} \text{ J})$	C1
	1.00 mol of deuterium forms 0.500 mol of helium-3	C1
	total energy $= 0.500 \times 6.02 \times 10^{23} \times 5.24 \times 10^{-13}$ $= 1.58 \times 10^{11} \text{ J}$	A1

56.

10(a)	spontaneous emission of (ionising) radiation	B1
	emission from unstable nucleus	B1
10(b)(i)	curve with decreasing negative gradient passing through $(0, N_0)$	B1
	curve passing through $(T, 0.5N_0)$	B1
	curve passing through $(2T, 0.25N_0)$ and $(3T, 0.125N_0)$	B1
10(b)(ii)	line through origin with positive gradient	B1
	straight line passing through (N_0, A_0)	B1
10(c)(i)	activity	B1
10(c)(ii)	decay constant	B1
10(d)	$N = N_0 \exp(-\ln 2 \times 1.70T / T)$	C1
	$N / N_0 = 0.31$	A1

57.

10(a)(i)	cannot predict when a (particular) nucleus will decay or cannot predict which nucleus will decay next	B1
10(a)(ii)	not affected by external / environmental factors	B1
10(b)(i)	line fluctuates or trend is a straight line	B1
10(b)(ii)	straight line of best fit drawn on Fig. 10.1	B1
10(b)(iii)	$M = M_0 \exp(-\lambda t)$ so $\ln M = \ln M_0 - \lambda t$ so gradient = $-\lambda$ (and magnitude of gradient = λ)	B1
10(b)(iv)	gradient = $(-)(8.0 - 4.8) / (11.6 - 0)$ (allow any correct pair of values from Fig. 10.1)	C1
	$\lambda = 0.28 \text{ s}^{-1}$	A1
10(b)(v)	half-life = $0.693 / \lambda$ = $0.693 / 0.28$ = 2.5 s	A1
10(c)	(for reaction to occur,) energy is released	B1
	energy release comes from fall in mass so total mass of products must be less (than mass of carbon-15)	B1

58.

8(a)	234, 92 for the uranium nucleus	B1
	4, 2 for the alpha particle	B1
8(b)(i)	$N_0 = 0.874 / (238 \times 1.66 \times 10^{-27})$ $= 2.21 \times 10^{24}$	A1
8(b)(ii)	$A = \lambda N$	C1
	$= \frac{\ln 2}{87.7 \times 365 \times 24 \times 3600} \times 2.21 \times 10^{24}$ $= 5.54 \times 10^{14} \text{ Bq}$	A1
8(b)(iii)	power $= 5.54 \times 10^{14} \times 5.59 \times 10^{-6} \times 1.60 \times 10^{-19}$	C1
	$= 496 \text{ W}$	A1
8(b)(iv)	$65.3 = 100e^{-\frac{\ln 2}{87.7} t}$	C1
	$\ln 0.653 = -(\ln 2 / 87.7) t$	A1
	$t = 53.9 \text{ years}$	
8(c)	advantage: less mass so less energy needed to launch probe	B1
	disadvantage: half-life shorter so will not provide power for as long	B1

59.

9(a)	time for activity (of sample) to halve	B1
9(b)	sketch: line with positive gradient starting at (0,0) and extending to $t = 80 \text{ min}$	B1
	exponential curve, extending from $t = 0$ to $t = 80 \text{ min}$, with gradient of steadily decreasing magnitude	B1
	line passing through (0,0), (20, $0.5N_0$) and (40, $0.75 N_0$)	B1
9(c)(i)	every (undecayed) nucleus has the same probability of decay	M1
	fewer (undecayed) nuclei remaining (with time), so fewer will decay (in a given time interval)	A1
9(c)(ii)	- sample emits in all directions but detector only captures emissions in one direction - some emissions are absorbed before reaching detector - some emissions are scattered within the sample - simultaneous arrival of multiple particles only registers once - some particles may reach detector but not cause ionisation <i>Any two points, 1 mark each</i>	B2
	measured count rate is less than the activity	B1

60.

9(a)	difference between mass of nucleus and (total) mass of nucleons	M1
	when infinitely separated	A1
9(b)(i)	neutron	B1
9(b)(ii)	$E = \Delta m c^2$	C1
	$\Delta m = (0.030377 - 0.002388 - 0.009105)u$ ($= 0.018884u$)	C1
	energy release $= (0.030377 - 0.002388 - 0.009105) \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2 = 2.8 \times 10^{-12} \text{ J}$	A1

9(c)(i)	number of atoms per unit time = $(1.4 \times 10^{28}) / (2.8 \times 10^{-12})$ $(= 5.0 \times 10^{39} \text{ s}^{-1})$	C1
	mass of one atom = $4 \times 1.66 \times 10^{-27}$ or $(4 \times 10^{-3}) / (6.02 \times 10^{23})$ $(= 6.64 \times 10^{-27} \text{ kg})$	C1
	mass per unit time = $6.64 \times 10^{-27} \times 5.0 \times 10^{39}$ $= 3.3 \times 10^{13} \text{ kg s}^{-1}$	A1
9(c)(ii)	$L = 4\pi\sigma r^2 T^4$	C1
	$1.4 \times 10^{28} = 4\pi \times 5.67 \times 10^{-8} \times (2.3 \times 10^9)^2 \times T^4$	
	$T = 7800 \text{ K}$	A1

61.

9(a)	(two small) nuclei join together	M1
	to form one larger nucleus	A1
9(b)	line with a peak at $A \approx 56$	B1
	line with steep initial positive gradient on the left of peak and shallower negative gradient at all points to the right of peak and line does not return to 0 binding energy	B1
9(c)(i)	X shown at value of A to the right of the peak	B1
9(c)(ii)	Y shown at value of A close to 1	B1
9(d)	energy from 1 nucleus = $(1.77 \times 10^{13}) / (6.02 \times 10^{23})$ $(= 2.94 \times 10^{-11} \text{ J})$	C1
	binding energy of Z = $[(1.25 + 1.81) \times 10^{-10}] - 2.94 \times 10^{-11}$ $(= 2.77 \times 10^{-10} \text{ J})$	C1
	nucleon number of Z = $93 + 139 + 2 - 1$ $(= 233)$	C1
	binding energy per nucleon = $(2.77 \times 10^{-10}) / (233 \times 1.60 \times 10^{-13})$ $= 7.43 \text{ MeV}$	A1